



Report

Nara Avila Moraes

Physics Institute
IFUSP

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Summary

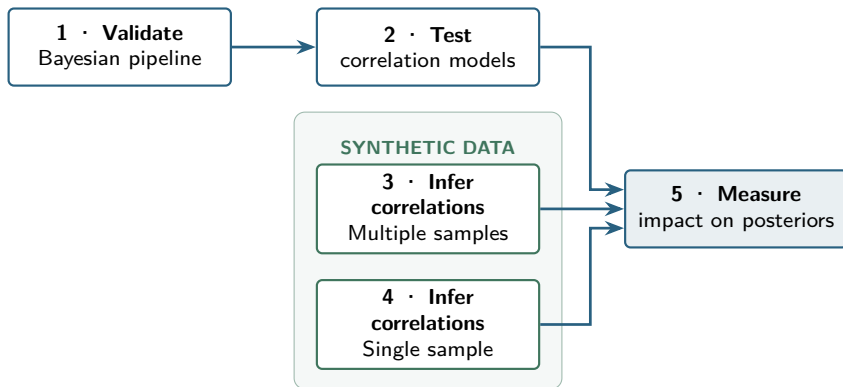
① Tasks Status

② Selected Article

Conference Materials Status

Task	Status	
2026-10 Conference	○	Simpósio IFUSP Presentation
2026-10 Conference	○	AGORÁ Presentation
2026-11 Poster	○	3rd School QC Poster
2026-12 Conference	○	LASNPA Presentation
2026-12 Paper	○	LASNPA Proceedings
2027-03 Conference	○	Quark Matter Presentation

Study Phases – Dependencies



Phase 1 – Bayesian Pipeline Validation

Task	Status	Output
↪ Reproduce Mankolli et al. results with the original JETSCAPE emulator	✓	Reference
↪ Reproduce the same results with the <code>hic_bayes</code> emulator	✓	Validation
↪ Emulator closure test	✓	Validation
↪ MCMC convergence test	✓	Diagnostics

Phase 2 – Correlation Model Sensitivity

Task	Status	Output
↪ Au–Au-200 + d–Au-200: Correlation-model sensitivity • Global CS correlation • Global RBF correlation • Global RBF correlation fix unit • Detector-grouped CS correlation • Detector-grouped RBF correlation	✓ ✓ 🕒 🕒 🕒	Posteriors
↪ Compare posterior shifts, credible intervals, and parameter correlations • Global correlation models • Detector-grouped correlation models	✓ 🕒	Analysis
↪ Quantify and compare the emulator-error and experimental-error covariance matrices in the likelihood	🕒	Analysis

Phase 3 – Correlation Inference from Multiple Samples

Task	Status	Output
↪ Generate controlled synthetic datasets with known covariance (N samples; goal: $N = 10$)	🌀	Dataset
↪ Classical correlation inference	○	Benchmark
↪ Quantum-learning correlation inference	○	Benchmark
↪ Classical vs. quantum comparison against the known covariance	○	Comparison

Phase 4 – Single-Sample Correlation Inference

Task	Status	Output
↪ Study correlation approximations when only one experimental sample is available		Heuristics
↪ Use reported uncertainty components to constrain plausible correlation structures		Models

Phase 5 – End-to-End Bayesian Impact

Task	Status	Output
↪ Propagate inferred covariance matrices through the Bayesian likelihood • Single sample: CS, RBF, Zero, GP • Multiple samples: Classical, Quantum	🔄	Posteriors
↪ Quantify covariance-estimation error → posterior error	○	Final study

Selected Article

Thermodynamic and Transport Properties of Quark-Gluon Plasma at Finite Chemical Potential with a DNN framework

Rishabh Kumar Tiwari¹, Kangkan Goswami¹, Suraj Prasad^{1,2}, Captain

R. Singh¹, Raghunath Sahoo^{1,*}, and Mohammad Yousuf Jamal³

¹*Department of Physics, Indian Institute of Technology Indore, Simrol, Indore 453552, India*

²*HUN-REN Wigner Research Center for Physics, Budapest 1121, Hungary and*

³*Key Laboratory of Quark and Lepton Physics (MOE) & Institute of Particle Physics,
Central China Normal University, Wuhan 430079, China*

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The characteristics of a thermal system depend strongly on its response to thermal gradients and the underlying microscopic interactions among constituents. In the present study, we investigate the thermodynamic and transport properties of the quark-gluon plasma (QGP) at finite baryon chemical potential within a deep-learning-assisted quasi-particle model (DLQPM). The temperature (T) and baryon chemical potential (μ_B)-dependent thermal masses of quasi-particles are estimated using neural networks trained to reproduce lattice QCD (lQCD) results for the equation of state, obtained via a Taylor-like expansion around vanishing baryon chemical potential. The trained model acts as an effective emulator, enabling us to estimate the thermodynamic and transport properties at finite μ_B . We compute the speed of sound, specific heat, viscosity, and conductivity of the deconfined medium. Our findings are in good agreement with available lattice calculations and other phenomenological models. The present study demonstrates that a DNN-based approach provides an efficient framework for studying the properties of the QGP at finite baryon density.

Figure: Alternative methods to infer transport parameters of QGP